SparseRank: Explainable Cost-Aware Feature Selection for Anomaly Detection in E-Commerce Microservices

Omar Bin Kasim Bhuian[1], Sang-Hoon Choi[2], M. A. Murad[3], Md Tanzimul Islam Refat[4], and Ki-Woong Park[*,5]

[1] Sejong University, Seoul, South Korea
omarbinkasimsefat@gmail.com
[2] Sejong University, Seoul, South Korea
csh0052@gmail.com
[3] Daffodil International University, Savar, Dhaka-1216, Bangladesh
masrukahmedmurad@gmail.com
[4] Daffodil International University, Savar, Dhaka-1216, Bangladesh
mdtanzimulislamrefat@gmail.com
[5] Sejong University, Seoul 05006, South Korea
woongbak@sejong.ac.kr

Abstract

Modern e-commerce platforms run on microservice architectures that emits massive stream of system-level signals. Collecting, storing, and analysing all of them is expensive and crucial to identify the root cause in case of abnormal behaviour. We present SparseRank, a lightweight framework that presents which monitoring signals actually matter for anomaly detection in a microservice of E-Commerce business. SparseRank reduces monitored feature space by 94% while maintaining detection quality by combining sparse feature selection with an unsupervised anomaly detector and a feature-attribution layer tested on a real benchmark that covers fourteen fault situations across twelve services.

1 Introduction

Modern e-commerce platforms increasingly depend on microservice architectures-commonly spans ten to twenty loosely coupled services to achieve independent scalability and continuous delivery. Anomaly detection over such systems has been addressed by unsupervised methods. Monitoring costs and explainability remain significant challenges. Recent work on microservice fault localisation focuses on root-cause identification after anomaly confirmation[1]. Our work addresses an earlier, complementary question: how to perform reliable anomaly detection while minimising the number of metrics that must be monitored? We address this through sparse feature selection that is simultaneously performance-conscious and explainable.

2 Literature Review

Explainability for anomaly detection in cloud systems has been addressed by causal methods. Eadro [2] introduces an end-to-end troubleshooting framework using logs, metrics, and traces jointly. While these methods improve detection quality, none addresses monitoring cost and explainability.

3 Methodology

This paper presents SparseRank, a three-stage pipeline: (i) L1-regularised logistic regression to select a compact feature subset by penalising redundant metrics to exactly zero; (ii) IsolationForest trained exclusively on the selected features to generate per-window anomaly scores; and (iii) TreeSHAP attribution on a surrogate decision tree to rank selected features by their contribution.

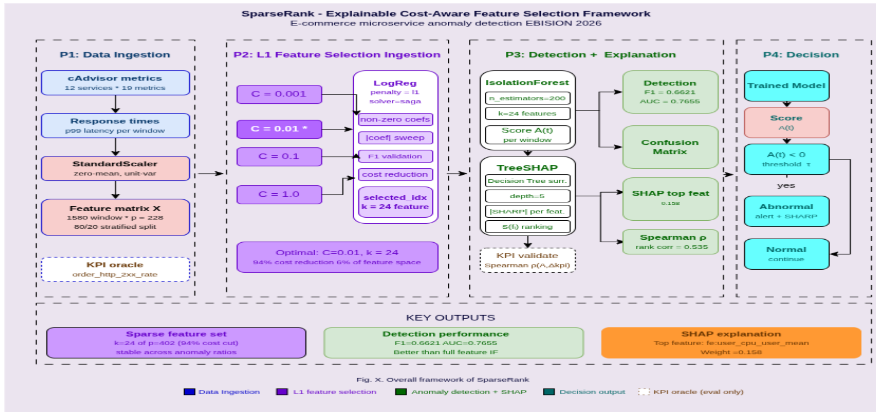

Figure 1: Overall framework of SparseRank.

4 Results

SparseRank achieved competitive performance compared with feature-selection baselines while reducing the monitored feature space. SparseRank identifies k = 24 from p=402 features, resulting in a 94% reduction in monitoring costs, while revealing the top contributing features through SHAP attribution.

5 Conclusion

We have demonstrated that SparseRank is a cost-effective, three-stage anomaly detection method. Ultimately, our proposed framework ensures cost optimisation in observability data monitoring using a dual-ended validation perspective combining dimensionality reduction and explanation-based interpretability. End-to-end Pipeline is available in a public GitHub repository.

6 Acknowledgments

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References

[1] Vajda, D. L., Do, T. V., Bérczes, T., and Farkas, K. "Machine Learning-Based Real-Time Anomaly Detection Using Data Pre-Processing in the Telemetry of Server Farms." *Scientific Reports*, 14(1), 23288, 2024.

[2] Lee, C., Yang, T., Chen, Z., Su, Y., and Lyu, M. R. "Eadro: An End-to-End Troubleshooting Framework for Microservices on Multi-Source Data." In *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*, pp. 1750–1762, IEEE, 2023.

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Abstract

Modern e-commerce platforms run on microservice architectures that emits massive stream of system-level signals — CPU, memory, network, disk, response time every second. Collecting, storing, and analysing all of them is expensive as well as it is crucial to identify the root cause in case of abnormal behaviour. We present SparseRank, a lightweight framework that presents which monitoring signals actually matter for anomaly detection in a microservice of E-Commerce business. SparseRank reduces monitoring costs by 94% while maintaining detection quality by combining sparse feature selection with an unsupervised anomaly detector and a feature-attribution layer. SparseRank demonstrates that careful preprocessing, a principled sparsity prior, and a model-agnostic explanation can all work together to provide a useful, deployable observability tool without the need for complex machinery when tested on a real benchmark that covers fourteen fault situations across twelve services.

Introduction

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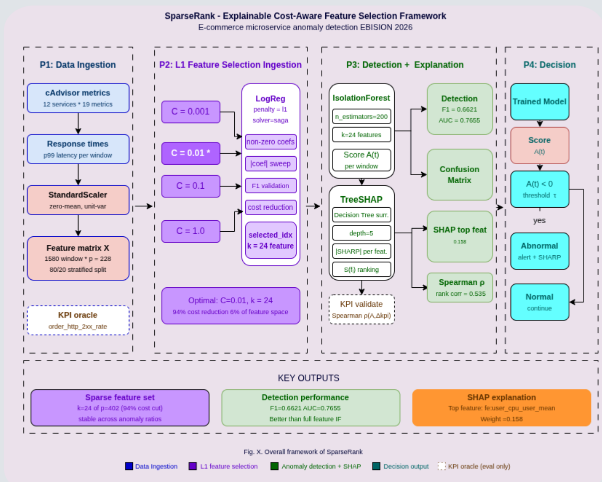

Figure 1: Overall framework of SparseRank.

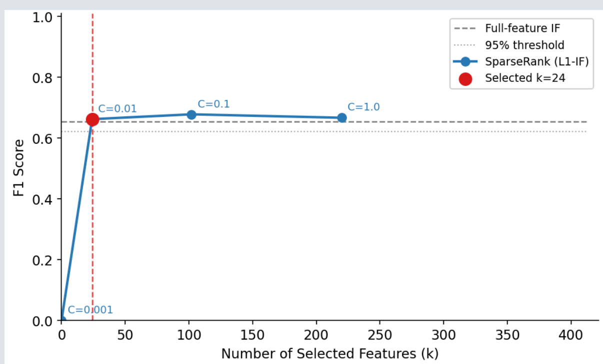

Figure 2: F1 feature Count trade off (L1 Sweep)

Literature Review

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Results

SparseRank achieved competitive performance compared with feature-selection baselines while reducing the monitored feature space. SparseRank identifies k = 24 from p=402 features, resulting in a 94% reduction in monitoring costs, while revealing the top contributing features through SHAP attribution.

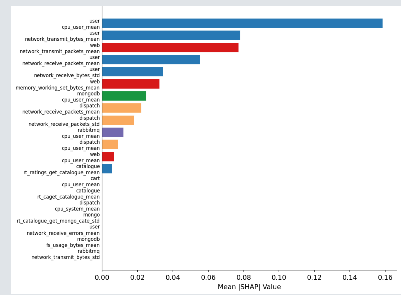

Figure 3: Top 20 SHAP feature Importances

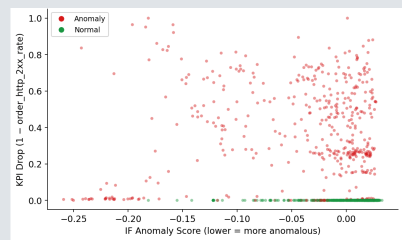

Figure 4: Anomaly Score vs Revenue KPI

Feature	Shap_mean_abs_sparse
user__cpu_user_mean	0.158430038641394
user__network_transmit_bytes_mean	0.0781166807575211
web__network_transmit_packets_mean	0.0771808662114891
user__network_receive_packets_mean	0.055235659669573
user__network_receive_bytes_std	0.0345305348257087
web__memory_working_set_bytes_mean	0.0325101285808732
mongodb__cpu_user_mean	0.0249199096098247

table 1: Shap feature importance

Conclusion

We have demonstrated that SparseRank is a cost-effective, three-stage anomaly detection method. Ultimately, our proposed framework ensures cost optimisation in observability data monitoring using a dual-ended validation perspective combining dimensionality reduction and explanation-based interpretability.

References

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